



Nivell 1

Crea una base de dades amb MongoDB utilitzant com a colleccions els arxius adjunts.

localhost:27017 > cinema

[Open MongoDB shell](#) [Create collection](#) [Refresh](#)

Collection name	Properties	Storage size	Documents	Avg. document size	Indexes	Total index size
comments	-	6.93 MB	50K	284.00 B	1	524.29 kB
movies	-	21.55 MB	24K	1.60 kB	1	364.54 kB
sessions	-	20.48 kB	1	540.00 B	1	20.48 kB
theaters	-	147.46 kB	1.6K	223.00 B	1	32.77 kB
users	-	36.86 kB	185	159.00 B	1	20.48 kB

Exercici 1

- Mostra els 2 primers comentaris que hi ha en la base de dades.
- Quants usuaris tenim registrats?
- Quants cinemes hi ha en l'estat de Califòrnia?
- Quin va ser el primer usuari/ària en registrar-se?
- Quantes pel·lícules de comèdia hi ha en la nostra base de dades?

Mostrar los dos primeros comentarios

En MongoDB Compass:

▼ Type a query: { field: 'value' } or [Generate query](#) ⚡

Project `{_id: 0, 'text': 1, 'date': 1 }`

Sort `{ "date": 1 }`

Collation `{ locale: 'simple' }`

Index Hint `{ field: -1 }`

Max Time MS

Skip Limit

EXPORT DATA ⓘ

100 1 - 2 of 2

```
{
  "text": "Optio totam dolores magni. Enim ratione fuga tempora voluptatum est cumque animi. Quia nam doloribus corrupti id voluptate esse. Tempore ma",
  "date": {
    "$date": "1970-01-01T01:07:09.000Z"
  }
}
```

```
{
  "text": "Dolorem animi tempora ullam quas. Iusto nobis reprehenderit aspernatur cupiditate a tempore. Commodi alias expedita dolore explicabo ipsam.",
  "date": {
    "$date": "1970-01-01T09:37:49.000Z"
  }
}
```

En Mongosh:

```
> db.comments.find(
  {},
  {
    '_id': 0,
    'text': 1,
    'date': 1
  },
  {
    'sort': { 'date': 1 },
    'limit': 2
  }
)
< {
  text: 'Optio totam dolores magni. Enim ratione fuga tempora voluptatum est cumque animi. Quia nam doloribus corrupti id voluptate esse. Tempore maiores o',
  date: 1970-01-01T01:07:09.000Z
}
{
  text: 'Dolorem animi tempora ullam quas. Iusto nobis reprehenderit aspernatur cupiditate a tempore. Commodi alias expedita dolore explicabo ipsam.',
  date: 1970-01-01T09:37:49.000Z
}
```

Cuántos usuarios hay registrados?

En MongoDB Compass:

Serían la cantidad de registros que hay en la base de datos USERS: 185 registros

localhost:27017 > cinema > users

Documents 185 Aggregations Schema Indexes 1 Validation

▼ Type a query: { field: 'value' } or [Generate query](#) ⚡

ADD DATA EXPORT DATA UPDATE DELETE

100 1 - 100 of 185

En Mongosh:

Por las dudas seleccione también que sean todos emails distintos

```
> db.users.countDocuments()  
< 185  
  
> db.users.distinct("email").length  
< 185  
  
cinema >
```

Cuántos cinemas hay en el estado de California?

Lo primero que hago es comprobar que California sea "CA" y que no haya repetidos ni mayúsculas o minúsculas.

```
> db.theaters.distinct('location.address.state')  
< [  
  'AK', 'AL', 'AR', 'AZ', 'CA', 'CO', 'CT',  
  'DC', 'DE', 'FL', 'GA', 'HI', 'IA', 'ID',  
  'IL', 'IN', 'KS', 'KY', 'LA', 'MA', 'MD',  
  'ME', 'MI', 'MN', 'MO', 'MS', 'MT', 'NC',  
  'ND', 'NE', 'NH', 'NJ', 'NM', 'NV', 'NY',  
  'OH', 'OK', 'OR', 'PA', 'PR', 'RI', 'SC',  
  'SD', 'TN', 'TX', 'UT', 'VA', 'VT', 'WA',  
  'WI', 'WV', 'WY'  
]
```

En MongoDB Compass:

Filtro state por "CA" y me devuelve que hay 169 registros de cines en California

Documents 1.6K Aggregations Schema Indexes 1 Validation

Query: { "location.address.state": "CA" }

Project: { field: 0 }

Sort: { field: -1 } or [['field', -1]]

Collation: { locale: 'simple' }

Index Hint: { field: -1 }

Buttons: Generate query, Explain, Reset, Find, </>, Options

Max Time MS: 60000

Skip: 0 **Limit:** 0

Actions: ADD DATA, EXPORT DATA, UPDATE, DELETE

Results: 25 1 - 25 of 169

```
{
  "_id": ObjectId('59a47286cfa9a3a73e51e72e'),
  "theaterId": 1008,
  "location": {
    "address": {
      "street1": "1621 E Monte Vista Ave",
      "city": "Vacaville",
      "state": "CA",
      "zipcode": "95688"
    },
    "geo": {
      "type": "Point",
      "coordinates": [
        -121.96328,
        38.367649
      ]
    }
  }
}
```

En Mongosh:

```
> db.theaters.countDocuments({'location.address.state': "CA"})
< 169
```

Quien es el primer usuario en registrarse?

En MongoDB Compass:

Ordeno por *id* porque los usuarios no tienen fecha de registro pero los *_id* tienen un 'timestamp' incluido. Uso el project para no mostrar información relevante de los usuarios.

Type a query: { field: 'value' } or Generate query

Project: { _id: 0, "name": 1 }

Sort: { _id: 1 }

Collation: { locale: 'simple' }

Index Hint: { field: -1 }

Buttons: Explain, Reset, Find, </>, Options

Max Time MS: 60000

Skip: 0 **Limit:** 1

Actions: EXPORT DATA

Results: 100 1 - 1 of 1

```
{
  "name": "Ned Stark"
}
```

En Mongosh:

```
> db.users.find({}, {_id: 0, 'name':1}).sort({_id: 1}).limit(1)
< {
  name: 'Ned Stark'
}
```

Cuántas películas de comedia hay en la base de datos?

Primero compruebo la lista de generos:

```
> db.movies.distinct("genres")
< [
  'Action',      'Adventure', 'Animation',
  'Biography',   'Comedy',    'Crime',
  'Documentary', 'Drama',     'Family',
  'Fantasy',     'Film-Noir', 'History',
  'Horror',      'Music',     'Musical',
  'Mystery',     'News',      'Romance',
  'Sci-Fi',      'Short',     'Sport',
  'Talk-Show',   'Thriller',  'War',
  'Western'
]
cinema>
```

En MongoDB Compass:

Filtro por Comedy y veo que hay 7024 registros.

The screenshot shows the MongoDB Compass interface. At the top, the query filter is set to `{'genres': 'Comedy'}`. The 'Find' button is highlighted. Below the query, the 'Project' field is set to `{ field: 0 }`, the 'Sort' field is set to `{ field: -1 } or [[ 'field', -1 ]]`, the 'Collation' field is set to `{ locale: 'simple' }`, and the 'Index Hint' field is set to `{ field: -1 }`. The 'Max Time MS' is set to 60000. The 'Skip' and 'Limit' fields are both set to 0. The interface shows 25 results per page, with 1-25 of 7024 records displayed. The first record is expanded, showing the following details:

- `_id`: ObjectId('573a1390f29313caabed4803')
- `plot`: "Cartoon figures announce, via comic strip balloons, that they will mov..."
- `genres`: Array (3)
 - 0: "Animation"
 - 1: "Short"
 - 2: "Comedy"
- `runtime`: 7
- `cast`: Array (1)

En Mongosh:

```
> db.movies.countDocuments({'genres': 'Comedy'})  
< 7024  
cinema>
```

Exercici 2

Mostra'm tots els documents de les pel·lícules produïdes en 1932, però que el gènere sigui drama o estiguin en francès.

En MongoDB Compass:

Utiligo \$GTE (grater than equal) y \$LT (less than) y luego el \$OR, ISODATE porque la fecha esta en ese formato.

The screenshot shows the MongoDB Compass interface. The query editor contains the following JSON query:

```
{  
  "released": {  
    "$gte": ISODate("1932-01-01"),  
    "$lt": ISODate("1933-01-01")  
  },  
  "$or": [  
    { "genres": "Drama" },  
    { "languages": "French" }  
  ]  
}
```

Below the query editor, the Project, Sort, Collation, and Index Hint settings are shown. The Project field is set to { field: 0 }. The Sort field is set to { field: -1 } or [['field', -1]]. The Collation field is set to { locale: 'simple' }. The Index Hint field is set to { field: -1 }. The Max Time MS is set to 60000. The Skip and Limit fields are both set to 0.

At the bottom, there is a table of movie documents. The table has 7 rows and 6 columns: _id, plot, genres, runtime, cast, and num_mflix. The first 6 rows contain movie data, and the 7th row is empty.

	_id ObjectId	plot String	genres Array	runtime Int32	cast Array	num_mflix
1	ObjectId('573a1392f29313...')	"A young French soldier ..."	[] 1 elements	76	[] 4 elements	2
2	ObjectId('573a1392f29313...')	"Famous motor-racing cha..."	[] 2 elements	85	[] 4 elements	1
3	ObjectId('573a1392f29313...')	"A tale of the love betw..."	[] 3 elements	80	[] 4 elements	1
4	ObjectId('573a1392f29313...')	"In this romance a libra..."	[] 2 elements	85	[] 4 elements	3
5	ObjectId('573a1392f29313...')	"A circus' beautiful tra..."	[] 2 elements	64	[] 4 elements	1
6	ObjectId('573a1392f29313...')	"A group of very differe..."	[] 2 elements	112	[] 4 elements	No field
7	ObjectId('573a1392f29313...')	"Wrongly convicted James..."	[] 3 elements	92	[] 4 elements	No field

En Mongosh:

```
> db.movies.find({
  released: {
    $gte: ISODate("1932-01-01"),
    $lt: ISODate("1933-01-01")
  },
  $or: [
    { genres: "Drama" },
    { languages: "French" }
  ]
})
< {
  _id: ObjectId('573a1392f29313caabcd99e3'),
  plot: 'A young French soldier in World War I is overcome with guilt when he kills a German soldier who, like himself, is a musically gifted conscript, ea
  genres: [
    'Drama'
  ],
  runtime: 76,
  cast: [
    'Lionel Barrymore',
    'Nancy Carroll',
    'Phillips Holmes',
    'Louise Carter'
  ],
  num_mflix_comments: 2,
  poster: 'https://m.media-amazon.com/images/M/MV5BYzJlMDZjM2QtNWU4Ny00ZGRiLTljNjctZGZjNGM2ZmIzN2I1XkEyXkFqcGdeQXVyMjUxODE0MDY@._V1_SY1000_SX677_AL_.jpg',
  title: 'Broken Lullaby',
  fullplot: "A young French soldier in World War I is overcome with guilt when he kills a German soldier who, like himself, is a musically gifted conscript
  languages: [
    'English'
```

Exercici 3

Mostra'm tots els documents de pel·lícules estatunidenques que tinguin entre 5 i 9 premis que van ser produïdes entre 2012 i 2014.

En MongoDB Compass:

Parecido al anterior

🕒

▼

```
{
  "countries": "USA",
  "awards.wins": { $gte: 5, $lte: 9 },
  "released": {
    $gte: ISODate("2012-01-01"),
    $lt: ISODate("2015-01-01")
  }
}
```

Generate query ⚙️

Explain

Reset

Find

⌕

Options ▶

➕ ADD DATA ▼

📄 EXPORT DATA ▼

✎ UPDATE

🗑 DELETE

💡

100 ▼

1 - 100 of 155

↺

↻

⌵

☰

|

{ }

🏠

🏠 movies						
	_id ObjectId	fullplot String	imdb Object	year Int32	plot String	genres Ar
1	ObjectId('573a13acf29313...	"The manager of the nega...	{ } 3 fields	2013	"When his job along with...	[] 3 eleme
2	ObjectId('573a13aef29313...	"When Caleb's gay roomma...	{ } 3 fields	2004	"After getting dumped by...	[] 3 eleme
3	ObjectId('573a13b5f29313...	"The Croods is a prehist...	{ } 3 fields	2013	"After their cave is des...	[] 3 eleme
4	ObjectId('573a13b9f29313...	"In 1938, the young girl...	{ } 3 fields	2013	"While subjected to the ...	[] 2 eleme
5	ObjectId('573a13b9f29313...	"Life for former United ...	{ } 3 fields	2013	"United Nations employee...	[] 3 eleme
6	ObjectId('573a13b9f29313...	"In 1999, the Janjira nu...	{ } 3 fields	2014	"The world is beset by t...	[] 3 eleme
7	ObjectId('573a13bbf29313...	"The story of the 1973 h...	{ } 3 fields	2013	"The story of the 1973 h...	[] 3 eleme
8	ObjectId('573a13bcf29313...	"In 1959, Alfred Hitchco...	{ } 3 fields	2012	"A love story between in...	[] 2 eleme
9	ObjectId('573a13bef29313...	"Marcus Luttrell, a Navy...	{ } 3 fields	2013	"Marcus Luttrell and his...	[] 3 eleme
10	ObjectId('573a13c1f29313...	"The true story of Paul ...	{ } 3 fields	2013	"The true story of Paul ...	[] 3 eleme
11	ObjectId('573a13c1f29313...	"An 83 year-old man retu...	{ } 3 fields	2009	"An 83 year-old man retu...	[] 2 eleme
12	ObjectId('573a13c1f29313...	"In the 1930s, an elderl...	{ } 3 fields	2013	"Native American warrior...	[] 3 eleme
13	ObjectId('573a13c1f29313...	"In the 1930s, an elderl...	{ } 3 fields	2013	"Native American warrior...	[] 3 eleme

En Mongosh:

```
> db.movies.find({
  countries: "USA",
  "awards.wins": { $gte: 5, $lte: 9 },
  released: {
    $gte: ISODate("2012-01-01"),
    $lt: ISODate("2015-01-01")
  }
})

< {
  _id: ObjectId('573a13acf29313caabd29366'),
  fullplot: "The manager of the negative assets sector of Life magazine, Walter Mitty, has been working for sixteen years for the magazine and has a tedious job.",
  imdb: {
    rating: 7.4,
    votes: 211230,
    id: 359950
  },
  year: 2013,
  plot: 'When his job along with that of his co-worker are threatened, Walter takes action in the real world embarking on a global journey that turns into a life-or-death struggle.',
  genres: [
    'Adventure',
    'Comedy',
    'Drama'
  ],
  rated: 'PG',
  metacritic: 54,
  title: 'The Secret Life of Walter Mitty',
}
```




Nivell 2

Exercici 1

Compte quants comentaris escriu un usuari/ària que utilitza "GAMEOFTHRON.ES" com a domini de correu electrònic.

En MongoDB Compass:

Se usa \ para que incluya el "." en la búsqueda. y finalmente /i para que no diferencie mayuscula de minuscula.

The screenshot shows the MongoDB Compass interface. The query filter is `{"email": "/gameofThron\\.es/i"}`. The options section includes: Project: `{ field: 0 }`, Sort: `{ field: -1 } or [['field', -1]]`, Collation: `{ locale: 'simple' }`, Index Hint: `{ field: -1 }`, Max Time MS: `60000`, Skip: `0`, and Limit: `0`. At the bottom, there are buttons for ADD DATA, EXPORT DATA, UPDATE, and DELETE, along with pagination controls showing 25 items per page and 1-25 of 22841 results.

En Mongosh:

```
> db.comments.countDocuments({'email': '/@gameOfThron\\.es/i' })
< 22841
```

Exercici 2

Quants cinemes hi ha en cada codi postal situats dins de l'estat Washington D. C. (DC)?

En MongoDB Compass:

Uso el Agregattions, primero filtro por DC agregando un stage llamado \$match

▼ Stage 1

\$match

```
1 {
2   "location.address.state": "WA"
3 }
4
```

Luego agrego otro stage llamado \$group, el *id* actua como groupby en Mongo, *total_cinemas* es el nombre que le doy al groupby, por cada valor le suma 1.

▼ Stage 2

\$group

```
1 {
2   _id: "$location.address.zipcode",
3   total_cinemas: { $sum: 1 }
4 }
5
```

Finalmente ejecuto el RUN

_id: "98409" total_cinemas : 2
_id: "98903" total_cinemas : 1
_id: "98502" total_cinemas : 1
_id: "98271" total_cinemas : 1
_id: "98188" total_cinemas : 2
_id: "98003" total_cinemas : 1
_id: "98373" total_cinemas : 1
_id: "98003" total_cinemas : 1

En Mongosh:

```

> db.theaters.aggregate([
  {
    $match: {"location.address.state": "WA" }
  },
  {
    $group: {_id: "$location.address.zipcode",
      total_cinemas: { $sum: 1 }
    }
  },
  {}
])

< {
  _id: '98383',
  total_cinemas: 1
}
{
  _id: '98036',
  total_cinemas: 1
}
{
  _id: '98005',
  total_cinemas: 1
}
{
  _id: '98516',
  total_cinemas: 1
}
{
  _id: '98027',

```



Nivell 3

Exercici 1

Troba totes les pel·lícules dirigides per John Landis amb una puntuació IMDb (Internet Movie Database) d'entre 7,5 i 8.

En MongoDB Compass:

Primero probe de manera normal y no me gusto como se mostraba

{

directors: ["John Landis"],

"imdb.rating": {\$gte: 7.5, \$lte: 8}

}

Generate query ↗

Explain

Reset

Find

</>

Options ▾

Project

{
 "_id": 0,
 "title": 1,
 "directors": 1,
 "year": 1,
 "imdb.rating": 1
}

Sort

{ }

Max Time MS

60000

Collation

{ locale: 'simple' }

Skip

0

Limit

0

Index Hint

{ field: -1 }

EXPORT DATA ▾

?

25 ▾

1 – 4 of 4

↺

↻

⌵

≡

| {}

🏠

🏠 movies

	imdb Object	year Int32	title String	directors Array
1	{ } 1 fields	1978	"Animal House"	[] 1 elements
2	{ } 1 fields	1980	"The Blues Brothers"	[] 1 elements
3	{ } 1 fields	1981	"An American Werewolf in..."	[] 1 elements
4	{ } 1 fields	1983	"Trading Places"	[] 1 elements

Luego utilice Aggregations para verlo mas bonito

localhost:27017 > cinema > movies

Documents 24K Aggregations Schema Indexes 1 Validation

Smatch Project Edit

Explain Export Run Options

ALL RESULTS Showing 1 – 4 count results

year : 1978
title : "Animal House"
director : "John Landis"
imdb_rating : 7.6

title : "The Blues Brothers"
year : 1980
director : "John Landis"
imdb_rating : 7.9

year : 1981
title : "An American Werewolf in London"
director : "John Landis"
imdb_rating : 7.6

title : "Trading Places"
year : 1983
director : "John Landis"
imdb_rating : 7.5

En Mongosh:

```

> db.movies.aggregate([
  {
    $match: {
      directors: "John Landis",
      "imdb.rating": { $gte: 7.5, $lte: 8 }
    }
  },
  {
    $project: {
      _id: 0,
      title: 1,
      director: "$directors",
      year: 1,
      imdb_rating: "$imdb.rating"
    }
  }
])
< {
  year: 1978,
  title: 'Animal House',
  director: [
    'John Landis'
  ],
  imdb_rating: 7.6
}
{
  title: 'The Blues Brothers',
  year: 1980,
  director: [

```

Exercici 2

Mostra en un mapa la ubicació de tots els teatres de la base de dades.

En MongoDB Compass:

Primero generar la proyección de las coordenadas.

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

</>

Options

Project

{

"location.geo.coordinates": 1}

}

Sort

{ field: -1 } or [['field', -1]]

Collation

{ locale: 'simple' }

Index Hint

{ field: -1 }

Max Time MS

60000

Skip

0

Limit

0

EXPORT DATA

251 - 25 of 1564

<

>

{ }

|

_id: ObjectId('59a47286cfa9a3a73e51e72c') location: Object geo: Object coordinates: Array (2) 0: -93.24565 1: 44.85466
_id: ObjectId('59a47286cfa9a3a73e51e72d') location: Object
_id: ObjectId('59a47286cfa9a3a73e51e72e') location: Object
id: ObjectId('59a47286cfa9a3a73e51e72f')

Segundo, exportar data en csv.



Export

Collection cinema.theaters

Export results from the query below

```
db.getCollection('theaters').find(  
  {},  
  { 'location.geo.coordinates': 1 }  
);
```

i Only projected fields will be exported. To export all fields, go back and leave the **Project** field empty.

Export File Type

JSON

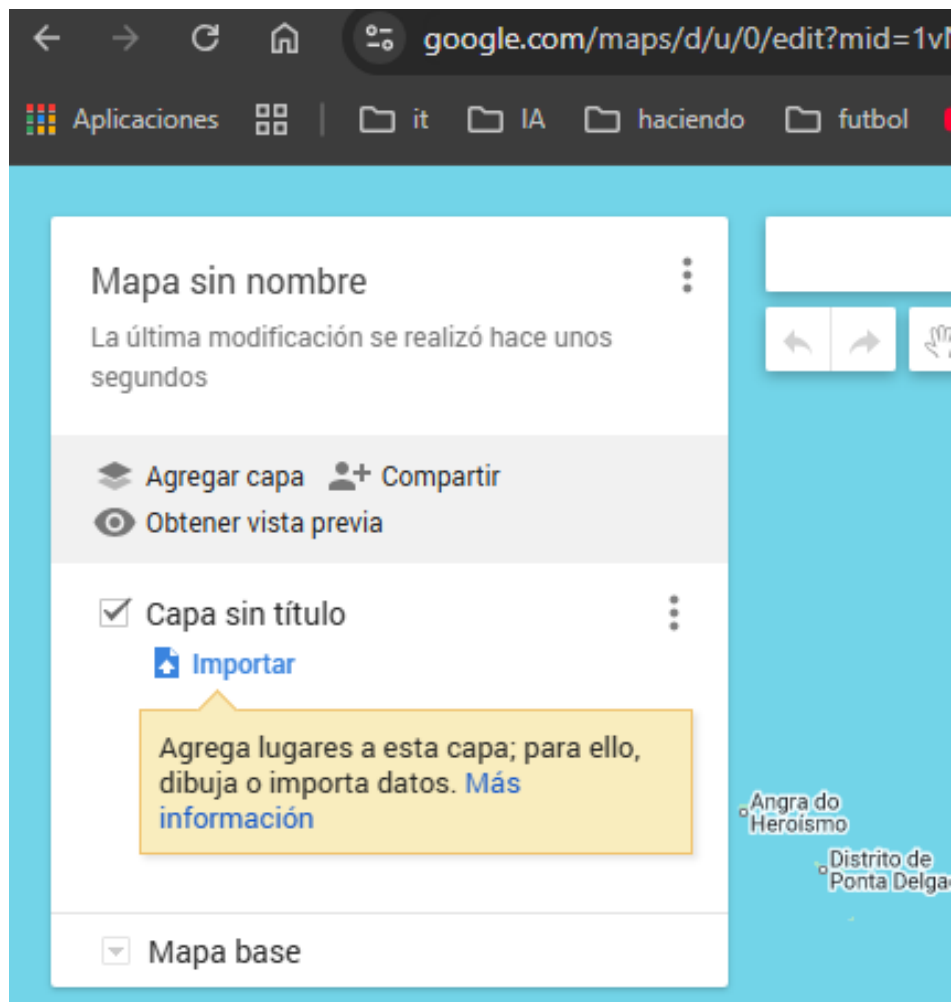
CSV

i Exporting with CSV may lose type information and is not suitable for backing up your data. [Learn more](#)

Cancel

Export...

Tercero, ir a google maps, crear un mapa e importar el archivo csv.



Cuarto, seleccionar las columnas que son Longitud y las que son Latitud

Selecciona columnas para colocar las marcas de posición

Selecciona las columnas de tu archivo que nos indican dónde se deben colocar las marcas de posición en el mapa como, por ejemplo, direcciones o pares de latitud y longitud. Se importarán todas las columnas.

☐ `_id` ?

☒ `location.geo.coordinates[0]`

☐ `location.geo.coordinates[1]`

Longitud

Latitud

Continuar Atrás Cancelar

Selecciona columnas para colocar las marcas de posición

Selecciona las columnas de tu archivo que nos indican dónde se deben colocar las marcas de posición en el mapa como, por ejemplo, direcciones o pares de latitud y longitud. Se importarán todas las columnas.

☐ `_id` ?

☒ `location.geo.coordinates[0]`

☒ `location.geo.coordinates[1]`

Longitud

Latitud

Continuar Atrás Cancelar

Quinto, seleccionar la columna que dara nombre a cada marca del mapa, en este caso solo tengo `_id` porque no tenian nombre los cines en nuestra base de datos.

Selecciona una columna como título para tus marcadores

Selecciona una columna que quieras utilizar como título para las marcas de posición como, por ejemplo, el nombre de la ubicación o de la persona.

☒ `_id` ?

☐ `location.geo.coordinates[0]` ?

☐ `location.geo.coordinates[1]` ?

Finalizar

Atrás

Cancelar

Finalmente se obtiene el mapa.

